

LECTURE 10

PANEL DATA MODELS (2) (CONT'D)

(A) Random Effects Model

(B) Correlated Random Effects

(C) IV Estimation in Panel Data Models

(A) Random Effects or Fixed Effects

- Which one should we use? It depends on whether a_i is correlated with the explanatory variables.
 - (i) If a_i is uncorrelated with all the explanatory variables, then the RE estimation is appropriate.
 - (ii) If a_i is correlated with some of the explanatory variables, then the FE (or first differenced) estimation is needed.
- How do we know whether a_i is uncorrelated with the explanatory variables, say $x_{it,j}$?
- Under the assumption that $Cov(x_{it,j}, v_{it}) = 0$, we could test if $Cov(x_{it,j}, a_i) = 0$.

- Recall: In the IV lecture, we use the Hausman test to test for endogeneity. The null hypothesis is $Cov(x, u) = 0$, i.e., both IV and OLS estimates are consistent.
 - ✓ We compare IV and OLS estimates.
 - ✓ If we reject the null hypothesis, then we use the IV estimators as they are always consistent (assuming that the instruments are valid)
- For the panel data regression, the null hypothesis is $Cov(x_{it,j}, a_i) = 0$.
 - ✓ We compare FE (consistent) and RE estimates (consistent when the null is true).
 - ✓ If we reject the null hypothesis, then we use the FE estimators as the a_i is removed and FE estimators are consistent.
- After FE and RE results are stored, we can run the Hausman test as follows.
 - > hausman FE name RE name

(B) Correlated Random Effects

- In the panel data regression,

$$y_{it} = \beta_0 + \beta_1 x_{it,1} + \cdots + \beta_k x_{it,k} + a_i + v_{it},$$

there are several ways to estimate parameters β 's: FD, FE and RE.

- First differenced and fixed Effects estimations:
 - Estimators are consistent for both cases: $Cov(x_{it,j}, a_i) \neq 0$ and $Cov(x_{it,j}, a_i) = 0$
 - We *cannot* to estimate the partial effects of time-invariant explanatory variables, e.g., race, gender, culture, etc.
- Random effects estimation:
 - Estimators are only consistent when $Cov(x_{it,j}, a_i) = 0$, otherwise inconsistent.
 - We *can* estimate the partial effects of time-invariant explanatory variables, e.g., e.g., race, gender, culture, etc.

- Is there a way that to estimate the coefficients of time-invariant explanatory variables when $Cov(x_{it,j}, a_i) \neq 0$?

✚ Yes. We can use correlated random effects (CRE) model. We *model* the correlation between a_i and x 's

- For simplicity's sake, assume the panel data regression model with only one x_{it}

$$y_{it} = \beta_0 + \beta_1 x_{it} + a_i + v_{it}. \quad (1)$$

The correlation between a_i and x_{it} can be modeled by regressing a_i on $\bar{x}_i$.

$$a_i = \gamma_0 + \gamma_1 \bar{x}_i + r_i, \quad (2)$$

where $\bar{x}_i = \frac{1}{T} \sum_t x_{it}$

Since r_i is the error of a_i after controlling for $\bar{x}_i$, we assume $Cov(\bar{x}_i, r_i) = 0$.

- With the projection of a_i on $\bar{x}_i$ in (2), the correlation between a_i and x_{it} depends on the value of γ_1 . $Cov(a_i, x_{it}) \neq 0$ whenever $\gamma_1 \neq 0$.

Substitute (2) $a_i = \gamma_0 + \gamma_1 \bar{x}_i + r_i$ in (1):

$$y_{it} = (\beta_0 + \gamma_0) + \beta_1 x_{it} + \gamma_1 \bar{x}_i + r_i + v_{it} \quad (3)$$

Since r_i is uncorrelated with x_{it} , we can use RE to estimate the equation (3). The equation (3) considers an additional important variable $\bar{x}_i$.

The RE estimators from this equation (3) are called the **correlated random effects** (CRE) estimators.

- CRE has the additional explanatory $\bar{x}_i$, to control for the correlation between a_i and x_{it} .

Remarks:

- (a) CRE $\neq$ RE because in the CRE we include $\bar{x}_i$
- (b) Interestingly, the estimated coefficients and their standard errors in CRE are same as that of FE. In other words, CRE of $\hat{\beta}_1 =$ FE of $\hat{\beta}_1$.
- (c) Multicollinearity in x_{it} and $\bar{x}_i$. can result in a bigger s.e. for CRE (= FE)
- (d) The CRE approach provides a simple formal way of choosing between the FE and RE approaches. We can conduct a t or F test of $H_0: \gamma_1 = 0$ against $H_1: \gamma_1 \neq 0$. If we reject H_0 at 5% level, we reject RE in favor of FE.

(C) IV Estimation in Panel Data Models

- In the panel data regression model:

$$y_{it} = \beta_0 + \beta_1 x_{it,1} + \cdots + \beta_k x_{it,k} + a_i + v_{it},$$

v_i could be correlated with $x_{it,1}, \dots, x_{it,k}$. This type of endogeneity renders OLS (and RE) inconsistent.

- When v_{it} is *uncorrelated* with explanatory variables, i.e., v_{it} is considered as pure random shocks, endogeneity due to a_i can be solved by FD or FE transformation.
- However, when v_{it} is *correlated* with explanatory variables, due to, e.g., simultaneity, after first-differencing or within transformation, endogeneity still exists.

Example:

Consider a simple panel data model $y_{it} = \beta_0 + \beta_1 x_{it} + a_i + v_{it}$. In addition, $Cov(x_{it}, a_i) \neq 0$ and $Cov(x_{it}, v_{it}) \neq 0$.

First differenced: $y_{it} - y_{i,t-1} = \beta_1(x_{it} - x_{i,t-1}) + (v_{it} - v_{i,t-1}) \quad (4)$

$$Cov(\textcolor{red}{x}_{it} - x_{i,t-1}, \textcolor{red}{v}_{it} - v_{i,t-1}) \neq 0$$

Within transformation: $y_{it} - \bar{y}_i = \beta_1(x_{it} - \bar{x}_{i.}) + (v_{it} - \bar{v}_i) \quad (5)$

$$Cov(\textcolor{red}{x}_{it} - \bar{x}_{i.}, \textcolor{red}{v}_{it} - \bar{v}_{i.}) \neq 0$$

- **Conclusion:** When $Cov(x_{it}, v_{it}) \neq 0$, FD and FE (CRE) are inconsistent. The Δv_{it} is correlated with the Δx_{it} (in FD) and $\ddot{y}_{it}$ (in FE).
- To deal with this type of endogeneity, we need instruments in first-differenced or within-transformed equation, and then run IV version of FD or FE regression.

- Consider a general panel data model,

$$y_{it} = \beta_0 + \beta_1 x_{it,1} + \cdots + \beta_k x_{it,k} + a_i + v_{it}$$

a_i can be removed using FD/FE transformation. The transformed model:

$$\ddot{y}_{it} = \beta_1 \ddot{x}_{it,1} + \cdots + \beta_k \ddot{x}_{it,k} + \ddot{v}_{it} \quad (6)$$

where $\dot{y}_{it} = y_{it} - y_{i,t-1}$ for FD, and $\ddot{y}_{it} = y_{it} - \bar{y}_i$ for FE transformation. $\ddot{x}_{it,1}$ and $\ddot{v}_{it}$ can be defined similarly. $\ddot{v}_{it}$ is correlated with, say $\ddot{x}_{it,1}$.

- The IV estimator of (6) is called the FD IV or FE IV estimator of β 's.
- For example, in the equation (4), the explanatory variable $x_{it} - x_{i,t-1}$ is endogenous. Suppose that z_{it} is a valid instrument for $x_{it} - x_{i,t-1}$. The IV estimator of β_1 is

$$\hat{\beta}_{1,IV} = \frac{\sum_{t=2}^T \sum_{i=1}^n (z_{it} - \bar{z})(y_{it} - y_{i,t-1})}{\sum_{t=2}^T \sum_{i=1}^n (z_{it} - \bar{z})(x_{it} - x_{i,t-1})}.$$

Stata command: xtivreg